

To: The Editors-in-Chief
Computer Physics Communications (CPC)
Elsevier Science
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Subject: Submission of Original Software and Computational Physics Manuscript
Title: *LeanFlow: A Neuro-Symbolic Architecture for Stiff Moduli Dynamics, Topological T-Duality, and Closed-Loop Anomaly Verification in String Cosmology*

Dear Editors of Computer Physics Communications,

We submit our original research manuscript for consideration as a Regular Article in *Computer Physics Communications*.

High-energy cosmological systems and string compactifications involve stiff, multi-scale systems of partial and ordinary differential equations. When scalar oscillation frequencies ($\omega \sim 10^3 H$) vastly exceed the cosmic expansion rate, classical explicit integrators experience catastrophic step-size collapse, while standard implicit solvers can drift into unphysically singular regions of moduli space. Concurrently, while interactive theorem provers (ITPs) like Lean 4 offer formal machine-checked proof certificates down to a minimal kernel, their interactive elaboration engines incur prohibitive interpretation latency (tens of milliseconds), historically precluding their use inside stiff numerical PDE integration loops.

In this work, we introduce **LeanFlow**: a neuro-symbolic framework that combines stiff numerical PDE integration with automated formal verification in Lean 4. A coupling to an external **rusty-SUNDIALS** (CVODE/IDA BDF) integration suite is planned but is not present in the submitted codebase, which loads that suite only from an absent sibling repository; the results below use SciPy’s Radau and `solve_bvp` solvers together with our own explicit Rust integrators.

The primary computational and algorithmic contributions of our work are:

- 1. Dual-Tier Execution Architecture:** LeanFlow separates invariant checking from numerical integration:
 - *Inner Loop (Rust Contract Assertions):* Metric positivity on the upper half-plane ($\tau_{\text{im}} > 0$), the Weak Energy Condition bound ($w \geq -1$), and Buscher-type reflection symmetry are treated as axiomatically assumed invariants that guide the integrator; they are compiled Ahead-of-Time into Rust SIMD contract assertions but are not kernel-verified in Lean 4 at the field level (metric, B -field, and dilaton). Per-residual overhead ($\approx 12\text{--}18\text{ ns}$) and stepping latency (0.12 ms) are estimates pending hardware verification; no benchmark harness for these figures currently exists in the repository.
 - *Outer Loop (Asynchronous Batch Gate):* Structural Lean 4 kernel checks (described below) are invoked asynchronously via non-blocking Unix domain socket IPC/CFFI ($\approx 45\text{ ms}$, pending hardware verification) at macroscopic simulation milestones, checkpoint boundaries, and topological defect extraction sweeps.
- 2. Native Duality Engine:** The integrator embeds continuous Buscher symmetry ($R \cdot \tilde{R} = \alpha'$) and modular group $SL(2, \mathbb{Z})$ invariance, axiomatically assumed rather than kernel-verified, as an operational numerical section condition. By co-simulating dual geometric frames and folding trajectories across the fundamental domain boundary $\mathcal{F}$, LeanFlow avoided coordinate singularities near self-dual points ($R \approx \sqrt{\alpha'}$) in the runs performed. A previously quoted metric minimum could not be traced to any output in the repository and has been withdrawn.

3. **Closed-Loop Neuro-Symbolic Triad:** We link numerical stochastic Langevin simulations to Topological Data Analysis (TDA) Mapper output (from the committed `tda_mapper_skeleton.js` 187 nodes, 557 edges, $\beta_1 = 376$ – the cycle rank of the Mapper graph, not a persistent-homology Betti number), partitioning defect configurations into candidate string-theoretic equivalence classes, and checking structurally, in the Lean 4 kernel, whether the resulting charges satisfy the algebraic Ramond–Ramond tadpole condition ($\sum Q_i = 0$). This check contains no `sorry` declarations, but it rests on Lean’s standard axioms together with additional project-specific axioms (not zero axioms), and its soundness depends on the fidelity of the upstream TDA extraction; we do not claim it certifies tadpole cancellation in the underlying physical theory.
4. **Preliminary Benchmark Estimates:** On the non-linear cosmological test equations described in the manuscript (Einstein–Klein–Gordon moduli flow, Mukhanov–Sasaki perturbations, and Symmetron screening BVP), we intend to benchmark LeanFlow against standard Python/SciPy BVP baselines and standard C++ SUNDIALS/CVODE, and to compare its cost on serverless spot GPU/TPU execution with scale-to-zero capability ($\text{min} = 0$) against dedicated HPC clusters. No benchmark execution log or timing harness exists in the repository, so we report no speedup or cost figures; the target figures in the manuscript’s comparison table are unmeasured.

All code, proofs, and simulation pipelines are open source. The codebase currently contains no `sorry` declarations; the Lean proofs rest on Lean’s standard axioms (`propext`, `Classical.choice`, `Quot.sound`) together with additional project-specific physics-domain axioms, and one local file (`proofs/FTheoryCosmology.lean`) is not a build target and does not compile. We withdraw a claim made in the original submission: the local `buscher_involution` result does not certify anything, because the underlying axiom (`inv_inv`), instantiated at `Nat`, proves `False`. Kernel-checked T-duality and lattice results we do rely on come from our LeanMaster foundation (release v2.2.0; theorems cited by name in the manuscript, depending only on Lean’s three standard axioms) and cover integer $O(d, d; \mathbb{Z})$ charge action and the generalized metric at $B = 0$, not the field-level Buscher rules asserted in the original letter.

This is a corrected resubmission. The original claims were deposited as Zenodo record 22683565 (DOI 10.5281/zenodo.22683565, concept 22683564); the corrected manuscript, with the withdrawn Buscher certification and hedged benchmark figures described above, will be issued as a new version of that concept record.

We believe LeanFlow, evaluated honestly against these caveats, remains of interest to the readers of CPC as a case study in combining stiff numerical integration with structural formal checks.

Thank you for your consideration.

Sincerely,

The Authors

SocrateAI Research Initiative